GENERAL ANATOMY

EPIPHYSIS

- The ends and tips of a bone which ossify from secondary centre are called epiphysis.
- **Classification:-**

§ According to number of epiphysis:-

SIMPLE
- ends of long bones develop from many epiphysis, which fuse independently with shaft.
e.g. femur.

COMPOUND
- ends of bones develop from many centre which unite to form single epiphysis which in turn fuses with shaft.
e.g. humerus.

§ Based on function:-

PRESSURE
- Articular
- Takes part in weight transmission
e.g. head of Humerus.

TRACTION
- Non articular
- Provide attachment to one or more tendon which exert traction on epiphysis.
e.g. Trochanter of femur

Grip pulling force

Ancestral

ATAVISTIC
- phylogenitically independent bone which in man becomes fused to another.
e.g. coracoid process of scapula.

ABERRANT
- not always presents
- e.g. at head of first metacarpal

Abnormal

COMPOUND
- In some bones 2-3 smaller epiphysis join to form compound epiphysis before joins a shaft.
e.g. Upper end of humerus.

BONES

- Bone is one-third connective tissue.
- It is impregnated which calcium constituting remaining two-third.
- **Classification:-**

§ According to shape:-

- LONG BONES - has elongated shaft (diaphysis) and two expanded ends (epiphysis) which are smooth & articular.
  - Typical long bones- two secondary epiphysis.
  - Miniature/short long bone - one epiphysis.
  - Modified long bones - no medullary cavity.

- SHORT BONES - shape is usually cuboid/scaphoid.
- FLAT BONES - form boundaries of certain body cavities.
- IRREGULAR BONES - Hip bone.
- PNEUMATIC BONES - Irregular bone with large air space.
- SESAMOID BONES - Bony nodule embedded in tendon/joint capsules.
- ACCESSORY BONES - Not always present. May occur as ununited Epiphysis developed from extra centre of ossification.

§ Development classification:-

- MEMBRANE (DERMAL) BONES - Ossify in membrane.
- CARTILAGINOUS BONES - Ossify in catilage.
- MEMBRANO-CARTILAGINOUS BONES - Ossify partly in membrane and partly in cartilage.
- SOMATIC BONES - Most bones are somatic.
- VISCERAL BONES - few and develop from pharyngeal arch.

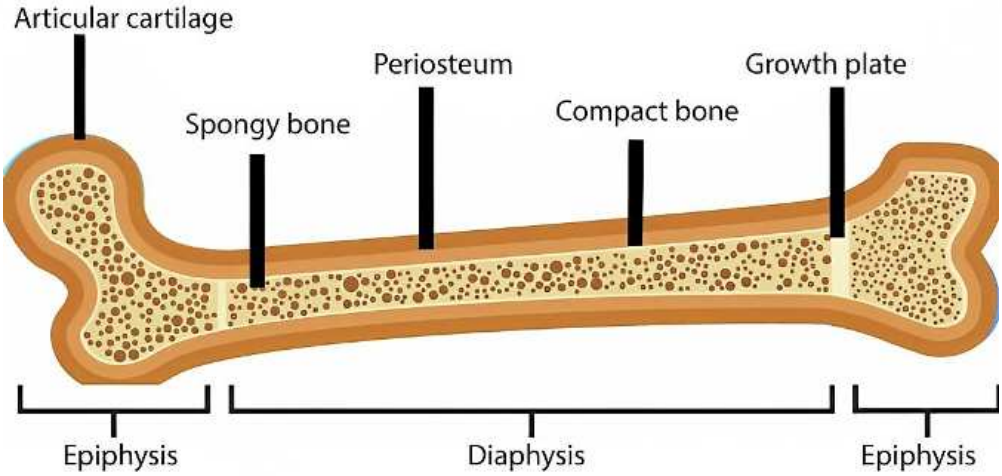
§ Regional Classification:-

- AXIAL SKELETON - includes skull, vertebrae, thoracic cage.
- APPENDICULAR SKELETON - Includes bone of limb.

§ Structural Classification:-

- MICROSCOPICALLY → COMPACT - bone is dense, texture is ivory but extremely porous.
- CANCELLOUS - Spongy or trabaecular bone is open in texture and is made up of meshwork of trabeculae.
- MICROSCOPICALLY → LAMELLAR - most of mature bones are composed of thin plates of bony tissue called lamallae.
- WOVEN - seen in fetal bones fracture repair or cancer bone.
- FIBROUS - young foetal bone.
- DENTINE, CEMENT - occur in teeth.

CC Lokesh Will Fuck DC



BLOOD SUPPLY OF LONG BONE

♦ **Arterial supply is derived from-**

§ NUTRIENT ARTERY-

Enters cortex through nutrient foramen → divides into ascending and descending branch → at the end, bend to form hairpin loop.

- Supplies metaphysis, inner 2/3rd cortex, medullary cavity.
- Direction of nutrient foramen is away from growing end, i.e., towards elbow away from knee.
- any blood infection causes emboli can block nutrient artery at bend and cause osteomyelitis.

§ PERIOSTEAL ARTERY-

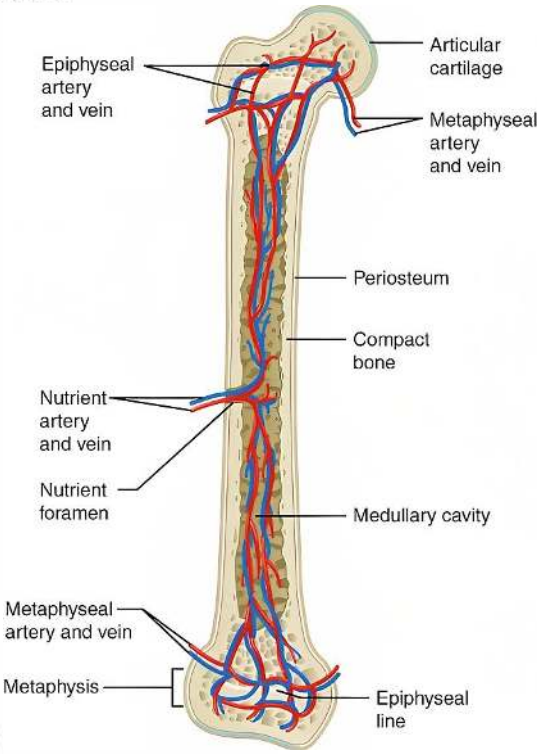
- numerous beneath mascular and ligamentous attachments.
- enter volkmann's canal to supply outer 1/3rd cortex.

§ EPIPHYSEAL ARTERY-

- derived from periarticular vascular arcades found on non-articular bony surfaces

§ METAPHYSEAL ARTERY-

- derived from neighbouring systemic vessels.
- pass directly into metaphysis and reinforce metaphysial branch from primary nutrient artery.



LONG SHORT BONE

♦ Nutrient Artery → enters middle of shaft → divide to form a plexus.
♦ major part of bone supplied by periosteal artery.

SHORT BONE

♦ Supplied by numerous periosteal arteries which enter their non- articular surface.

VERTEBRA

♦ Body - supplied by anterior posterior vessels.
♦ Vertebral arch - Supplied by large vessels entering base of transverse process.

RIBS

♦ Nutrient artery - enter just beyond tubercle.
♦ periosteal artery

VENOUS DRAINAGE -

♦ Veins are large and numerous - cancellous bone.
♦ Accompany artery in Volkmann's canal - compact bone.

CARTILAGE

• Cartilage is connective tissue composed of cells and fibres.
embedded in a firm, get. like matrix which is rich in a mucopolysaccharide.

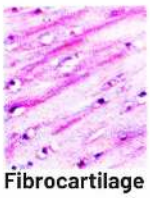
♦ Classification → ELASTIC
→ HYALINE
→ FIBROCARTILAGE

§ HYALINE -


Hyaline Cartilage

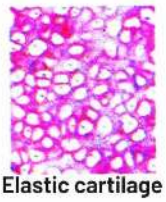
- Located in articular cartilage of long bone,epiphyseal plates, nasal cartilage, thyroid, trachae, bronchi, coastal cartilage.
- Bluish white in colour.
- Shiny or translucent.
- Very thin fibres.
- flexible.
- perichondrium is present.

§ FIBROCARTILAGE -


Fibrocartilage

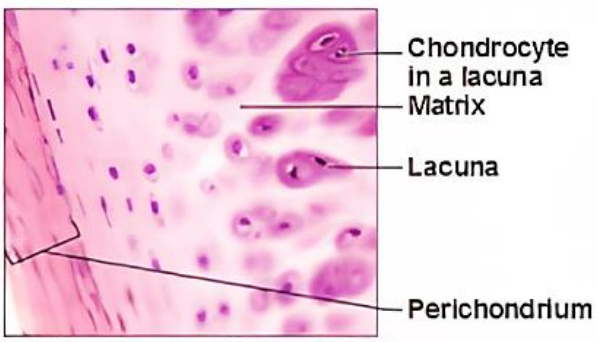
- Located in intervertebral disc, Temporomandibular joint, stenoclavicular joint.
- Glistening white in colour.
- Opaque appearances.
- Numerous white fibres.
- perichondrium is absent.

§ ELASTIC -

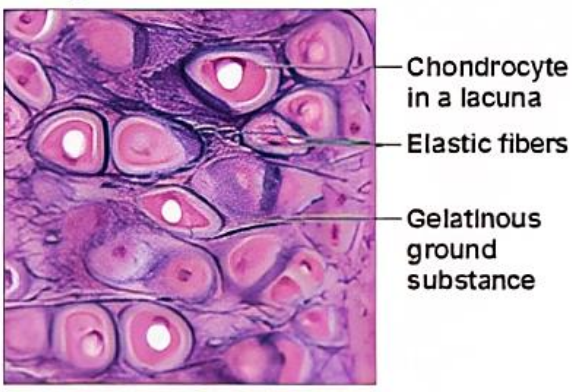

Elastic cartilage

- located in pinna, external auditory meatus, eustachian tube, epiglottis.
- yellowish in colour.
- opaque appearance.
- numerous yellow fibres.
- Most flexible.
- perichondrium is present.

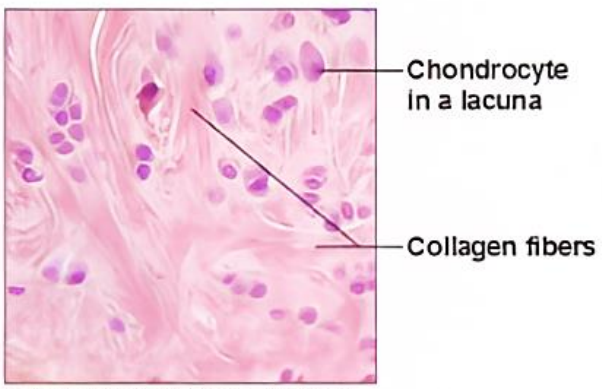
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(a) Hyaline cartilage (180×)



(b) Elastic cartilage (470×)



(c) Fibrocartilage (285×)

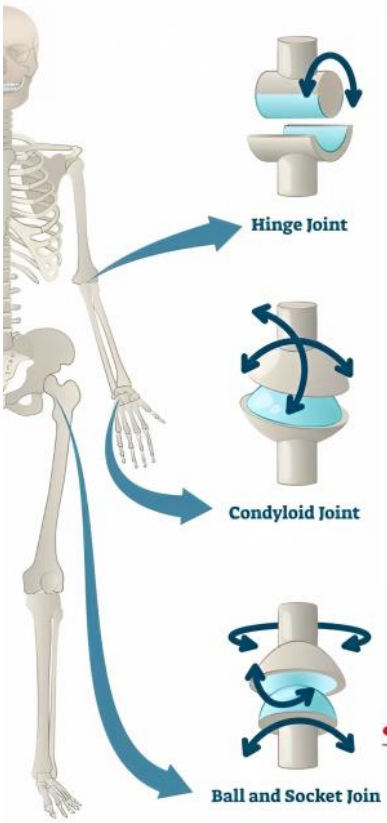
JOINT

- It is a junction between two or more cartilage.
- Device to permitt movement.

§ Classification -

• Structural Classification -

- **FIBROUS JOINTS** — Immovable or permit slight movement
  - SUTURES - present only in skull.
  - SYNDESMOSIS - fibrous union between bones.
  - GOMPHOSIS - peg and socket junction between tooth and socket.
- **CATILAGINOUS JOINTS** - bones are joined by plates of cartilage.
  - PRIMARY/SYNCHONDROSIS - temporary in nature as after a certain age cartilage plate is replaced by bone.
  - SECONDARY/SYMPHASES - permanent and persist throughout life.
- **SYNOVIAL JOINTS** - most evolved - most mobile.
  - BALL AND SOCKET JOINT
  - SELLAR / SADDLE JOINT
  - CONDYLAR/BICONDYLAR JOINT
  - ELIPSOID JOINT
  - HINGE JOINT
  - PIVOT JOINT
  - PLANE JOINT



• Functional Classification -

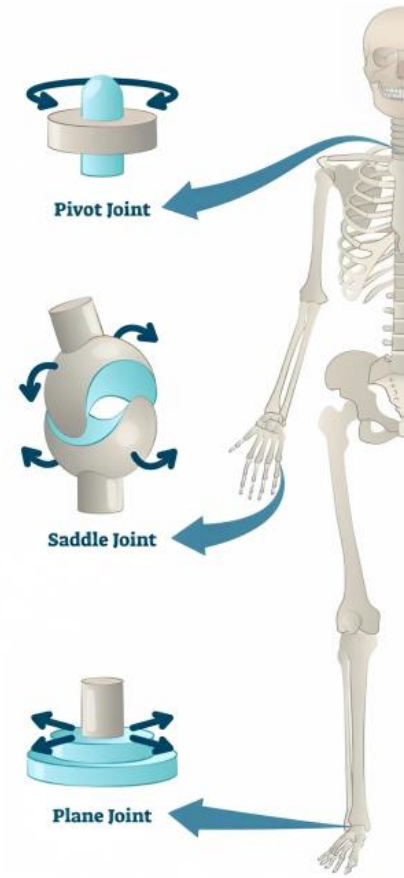
- SYNARTHROSES - Immovable
- AMPHIARTHROSES - Slightly movable.
- DIARTHROSES - freely movable.

• Regional Classification -

- SKULL TYPE - Immovable
- VERTEBRAL TYPE - Slightly movable.
- LIMB TYPE - freely movable.

• According to number of articulating bones -

- SIMPLE JOINT - When only two bones articulates.
- COMPOUND JOINT - More than two bones articulate in capsule.
- LIMB TYPE - when joint cavity is divided by an intra-articular disc.

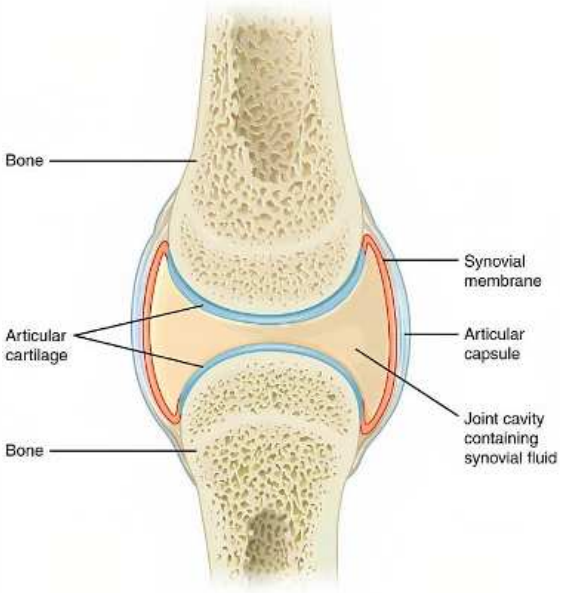


SYNOVIAL JOINT

♦ most evolved and most mobile.

♦ **Characters:-**

- Articular surface are covered with hyaline cartilage.
- Synovial fluid is filled between articular surfaces.
- Joint cavity is surrounded by articular capsule.
- Varying degree of movement are always permitted.



♦ **Classification:-**

1. PLANE SYNOVIAL JOINT -

- Articular surfaces are more or less flat.
- Permit gliding movement in various directions.

articular cartilage

Eg. § Intercarpal joints. § Intertarsal joints.
§ Carpometacarpal (except first).
§ Tarsometatarsal § Intermetatarsal

2. HINGE JOINT -

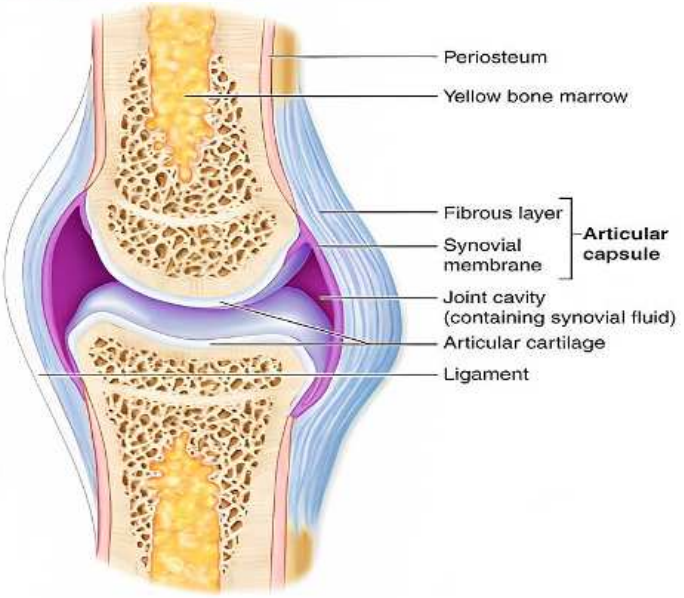
- Articular surfaces are pulley shape.
- Movements are permitted in one plane.

Eg. § Elbow joint. § Ankle joint.

3. PIVOT JOINT -

- Comparises a control bony pivot surrounded by osteoligamentous ring.
- Movement permitted in one plane.

Eg. § Median atlantoxial joint.
§ Superior and inferior Radioulnar joint.



4. CONDYLAR JOINT -

- Articular surface includes two distinct condyles.

Eg. § Knee joint. § TMJ.

5. ELLIPSOID JOINT -

- Articular surface include an oval.
- Free movements are permitted.

Eg. § wrist joint. § Atlanto - axial joint.

6. SADDLE JOINT -

- Articular surface are reciprocally concavo convex.
- Movements are similar to these permitted by an ellipsoid joint.

Eg. § Sternoclavicular joint. § first carpometacarpal joint.

7. BALL-SOCKET JOINT -

- Articular surfaces include a globular head fitting into a cup-shaped socket.
- Movements occur around indefinite axis.

Eg. § Shoulder joint. § Hip joint.

KARYOTYPING

♦Also called 'IDIOGRAM'.

♦ Standard arrangement of photographed metaphase spread of chromosome in decreasing length is called 'KARYOTYPING'.

♦ Chromosomes are identified on the basis of:-

- Length of arm of chromosome.
- Position of centromere of chromosome.

Procedure -

- done by culture of lymphocyte in medium of phytohaemagglutinin (PHA).

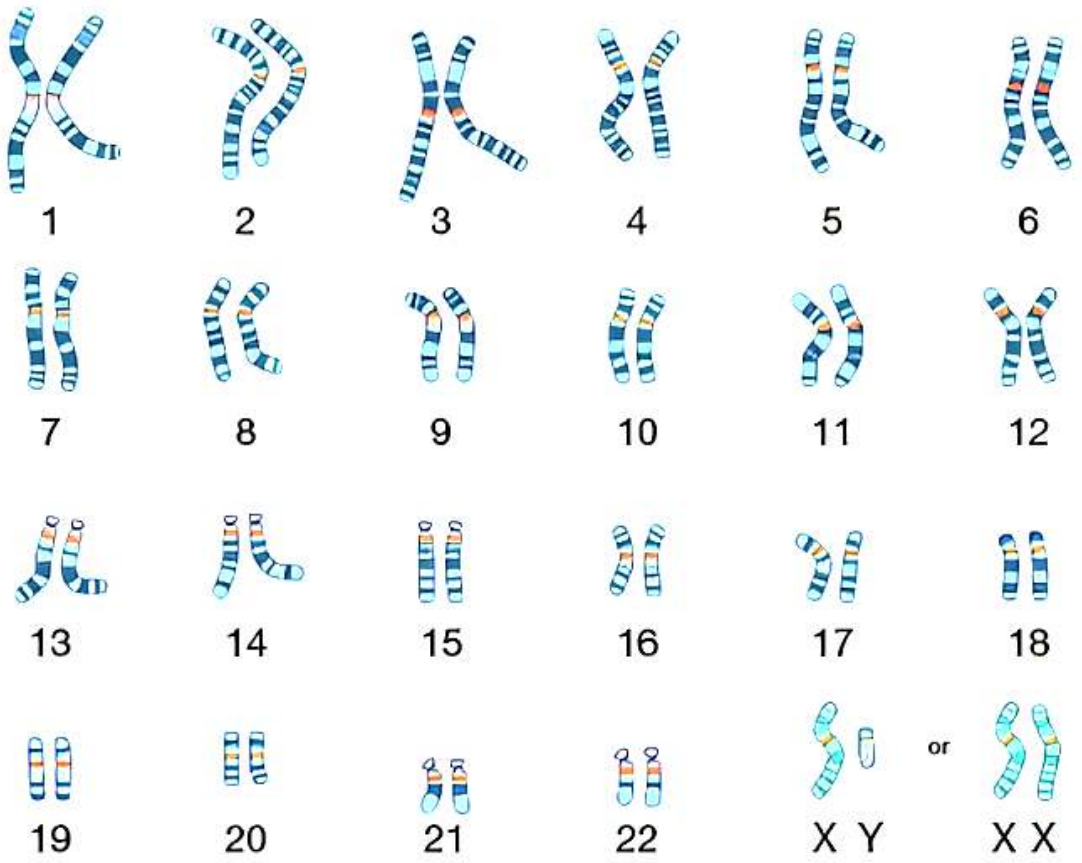
Cell divison → Arrested at metaphase by action of colchicine → Chromosome are spread, counted and photographed under microscope → image are cut and arranged in pair in decreasing order of length. → Chromosomes are numbered 1-22 in decreasing order of length.

Karyotyping helps to:-

1. Identifying pattern of abnormal chromosome.
2. Determine the sex.

♦ The length of pair number 1 is long and 22 is short.

♦ Individual chromosomes can be recognized with quinacrine band and Giemsa band.



DOWNS SYNDROME

♦ Most common congenital anomaly due to numerical abbreation first described by Dr. Down.
♦ There is trisomy of 21st chromosome, i.e., 47xx or 47xy.
♦ Commonly seen in early primigrivida or infants born to female suffering from viral infection during pregnancy.

Clinical features :- • Mentally retarded. • Poor growth and short stature. • Widely spread great toe.

Facial features :- • Moon face/short head/epicanthal fold/narrow palate/protruding tongue/ flat naval bridge/small ears.

Hand features :- • Short and broad hand / single characterstic crease called siminean crease.

Prenatal diagnosis :- • Aminocentesis • Increased Inhibin A level.

♦ Also called mongolism due to characterstic mongolian face of patient.

KLINEFELTER's SYNDROME

♦ It is sex-chromosome abnormality with an extra X chromosome in male.
♦ Also called xxy syndrome.
♦ Karyotype is 47 xxy.
♦ Barr body is seen in afflicted males.
♦ Occurs due to meiosis II nondisjunction.

Clinical features :- • small degree of mental retardation.
• Small testis leading to infertility.
• Small penis and scrotrum.
• Gynaecomastia.
• Scanty pubic and facial hairs.

TURNER's SYNDROME

♦ Sex chromosomal abnormality with partial or complete loss of X chromosome.
♦ Karyotype is 45x chromosome.
♦ No Barr body is seen in buccal smear as one chromosome is absent.

Clinical features :-

• Short stature • Webbed neck.
• Low set ears • normal/slightly retarded intelligence. • cubitus valgus, i.e. increased carrying angle at elbow.
• Amenorrhoea.
• Widely placed nipples.
• Rudimentary overy/streak gonads.

Treatment :- • Replacement therapy.
• IVF for sterile females.